

Motor Insurance Pricing with GLMs and GAMs

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1. Exploratory data analysis

1.1. Original Data Import and Cleaning

The dataset comprises 163 657 automobile insurance policies and originally contained 16 variables. We began by renaming and recoding for consistency: numeric variables now include ageph (policyholder age), expo (policy exposure in years), lnexpo (log-exposure), nclains (number of claims) and amount (total claim cost), while we derived avg = amount/nclains (set to NA when nclains = 0). All categorical fields—postal code (codposs), coverage type, fuel type, gender, vehicle use, fleet indicator, sports car flag, claim-count split, car age band, and power band—were converted to factors. We then merged a Belgian postal-code lookup table (inspost.xls) into the main file on codposs, retaining all 163 657 rows and flagging any unmatched codes for later spatial exclusion.

1.2. Claims data set

Response Distribution

We illustrate our methodology with data given from Belgium. Each record in the data set represents a unique policyholder who is observed during a certain policy period, ranging from one day to one year. The data set contains 163 657 policyholders, and the available variables are listed in Table 1.

Variable	Description
ageph	age of the policyholder
codposs	code in Belgium
duree	exposure, fraction of the year the insured is covered
lnexpo	log of exposure
nbrtotc	total number of claims reported during period of exposure
chargetot	total claim amount, aggregated over the nbrtotc reported claims
agecar	age of the car: 0 – 1, 2 – 5, 6 – 10, > 10
sexp	sex of the policyholder: male or female
fuelc	type of fuel: petrol or gasoil
split	split of the premium: monthly, once, twice, three times per year
usec	use of the car: private or professional
fleetc	car belonging to a fleet: yes or no
sportc	sport car: yes or no
powerc	power of the car: <66, 66-110, >110

Table 1: overview of the available variables

Figure 1 shows how the claim frequency, exposure and severity distribute. It illustrates that about 87% of the policyholders report no claims, while about 10 % have exactly one claim. Policies with two claims account for just 1 %, and three or more claims are exceedingly rare. The exposure is overwhelmingly dominated by one-year policies, nearly 80%, the other policyholders is equally

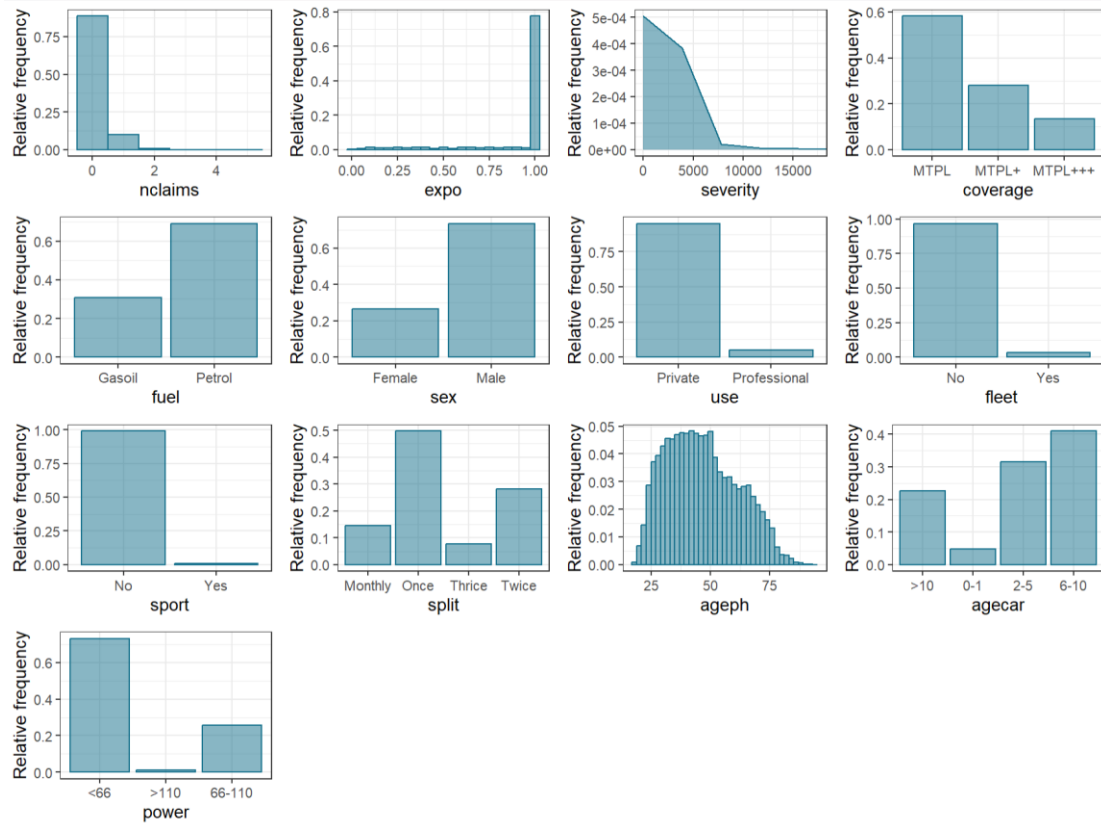
spread out between zero and one. Policyholders with an exposure lower than one have surrendered the policy during the year or started the policy in the course of the year. The overall claim frequency of the portfolio, calculated as the ratio of the total number of claims and the total exposure in years, is equal to 13.93%. Claims mainly involve small amounts. The overall claim severity of the portfolio, calculated as the ratio of the total claim amounts and the total number of claims, is equal to 1622.05 Euro.

Figure 1 also illustrates how the risk factors from Table 1 are distributed. The data set contains 9 categorical risk factors: age of the car, coverage, fuel, sex, use, fleet, split, sport of the car, power of the car. The portfolio comprises nine categorical risk factors. Car age is skewed toward middle-aged vehicles: 41.0 % are 6–10 years old, 31.5 % are 2–5 years, 22.7 % exceed 10 years and only 4.75 % are under one year. Coverage levels are 58.3 % basic MTPL, 28.2 % MTPL+ and 13.5 % full omnium. Fuel types split into 69.2 % petrol versus 30.8 % diesel, and policyholders are 73.6 % male and 26.4 % female. Use is predominantly private (95.1 % vs. 4.9 % professional), only 3.2 % of vehicles belong to fleets and 0.9 % are sports cars. Payment frequency is 49.8 % once, 28.1 % twice, 14.4 % monthly and 7.7 % thrice. Finally, engine power clusters below 66 kW (73.2 %), mid-range 66–110 kW (25.8 %) and above 110 kW (1.0 %).

The data set also contains only one continuous risk factor: ageph. The histogram (binwidth=2 years) shows almost all policyholders (93.53%) are aged between 25 and 75, which means that there are few young and old drivers in the data set. The 25th percentile is age 35, median age 45, and 75th percentile 58.

1.3. Spatial Exploration

Using the merged postal-code shapefile, we produced different maps of total exposure and of claim frequency by postal area, where the policyholder resides. The map of Belgium in Figure 2 visualizes the exposure in each municipality relative to the area of the municipality. These maps highlight geographic hotspots—regions where exposure is concentrated and where claim rates exceed national averages. The `1_spatial_EDA.R` script begins by reading in policy data (already loaded as `df`), converting postal codes to character and ensuring numeric fields (`expo`, `nclaims`, `amount`) are properly typed. Next, total exposure can be summarized by postal code and joined that back to the spatial object on matching codes. To visualize relative density, we compute exposure per unit area and classify it into three quantile-based bins (“low(20%),” “average,” “high(80%)”) while white municipalities are those where the insurer has no policyholders and therefore not exposed to the risk of filing a claim.



The resulting map highlights where policy exposure—and by extension, claim potential—is most concentrated. Darker blue areas cluster around Belgium’s major urban centers (Brussels, Antwerp and Ghent), reflecting high vehicle density and market penetration. In contrast, lighter or white regions correspond to rural zones with sparse exposure. This spatial pattern confirms that claim frequency models should account for geographic heterogeneity—urban postal codes will likely exhibit higher claim counts per area than their rural counterparts.

Claim Frequency by Postal Code Area

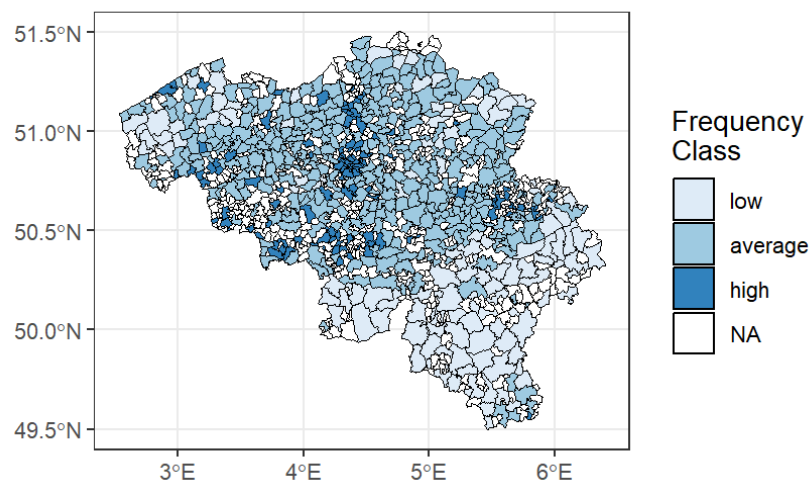


Figure 1

2. GLM Frequency Modelling

A random 80/20 split of the 163 657 policies created a training set and a hold-out test set. All models below were fitted on the training data, with the log-exposure (lnexpo) included as an offset.

2.1. GLM Frequency Modelling

A “full” Poisson generalized linear model (GLM) was first fitted using all ten predictors. To select variables, a drop in deviance analysis is used. 10 candidate model specifications g1-g10 were created (adding one or two predictors at a time) via likelihood-ratio (chi-square) tests. Every main effect proved highly significant, whereas the only two-way interaction tested (sex × fuel) did not improve fit ($p > 0.05$), so no interactions were retained. Next, predictors were then entered sequentially in the order coverp→fuelc→sexp→agecar→split→usec→fleetc→sportc→powerc→ageph. At each step, a chi-square test compared the model before and after adding the next variable. All variables up through split yielded significant deviance reductions ($p < 0.001$), but “use” (usec) did not, and was therefore dropped.

Starting from this nine-covariate model, a formal drop1(test="Chisq") was used to evaluate each term’s contribution when all others remain in the model. This confirmed that fleet membership (fleetc) and sport-car flag (sportc) did not significantly reduce residual deviance ($p > 0.10$) and were removed, yielding a final Poisson GLM with seven predictors: coverage, fuel, sex, age-band, split, power, and policyholder age.

Comparing AIC values showed that the reduced seven-term Poisson model attained a lower AIC than the full specification, confirming parsimony without loss of fit. A negative-binomial (NB) model was then fitted using the same predictor set; its AIC was lower, as Table 2 shows.

Model	df	AIC
Poisson	14	100768.8
NB	15	100470.6

Table 2 Comparison AIC of two models

Together, these results support the NB model—with coverage, fuel type, gender, car-age band, payment split, power band, and policyholder age—as the preferred frequency model for these data.

2.2. GLM Severity Modelling

A Gamma GLM with log link (and claim counts as weights) was fitted first, using all ten predictors. Although the overall fit was acceptable, several covariates showed no statistical significance ($p > 0.05$), suggesting misspecification of the Gamma assumption or overdispersion.

Next, we modeled log(sev_avg) via an ordinary Gaussian GLM (identity link on the log scale), again weighted by claim counts. This log-normal specification yielded a substantially lower AIC than the Gamma model, indicating better empirical fit.

To simplify, a drop-in-deviance (“drop1”, test = “F”) analysis was applied to the log-normal model. This revealed that fuel type, gender and power band did not contribute significantly once the other terms were included (all $p > 0.10$), so these three factors were removed. The reduced model—retaining only coverage level, car-age band, payment split and policyholder age—achieved an even lower AIC than the full log-normal fit, confirming that the parsimonious specification is preferable for severity.

Model	df	AIC
Log-normal (Full)	15	66146.64
Log-normal (Slimmed)	15	66141.75

Table 3

Using the final, pared-down frequency and severity models, we constructed a full “technical tariff”

schedule in Excel that assigns a pure premium factor to each combination of risk characteristics. We then applied these factors to the exposure in our data to produce aggregate predictions of claim counts and total claim costs for three samples: the full portfolio, the 80% training set, and the 20% hold-out test set.

Dataset	Actual Total Claims	Predicted Total Claims	Actual Total Paid	Predicted Total Paid
Full	20 290	20 264	32 911 482	28 383 713
Train	16 171	16 204	26 198 294	22 695 999
Test	4 119	4 060	6 713 188	5 687 714

Table 4

Across all three samples, predicted claim counts track actual counts to within about 1%, demonstrating that the frequency model generalizes well. Total paid amounts are systematically lower in the predictions—by roughly 10–15%—reflecting the log-normal severity model’s conservative treatment of extreme losses. Nonetheless, the tariff delivers accurate frequency forecasts and reasonable severity estimates, making it a robust basis for rate-making and reserve planning.

3. GAM Modelling

GAMs are a suitable tool for actuarial regression modeling because of their flexibility in handling different types of risk factors. In this dataset, only ageph is needed to be dealt with, by using mgcv package in R. Also, it allows for spatial risk factors, which showed in Figure 3. The predictor η of the GAMs is expressed as follows:

$$\eta_i = g(\mu_i) = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}^d + \sum_{j=1}^q f_j(x_{ij}^c) + \sum_{j=1}^r f_j(x_{ij}^s, y_{ij}^s).$$

3.1. GAM Frequency Modelling

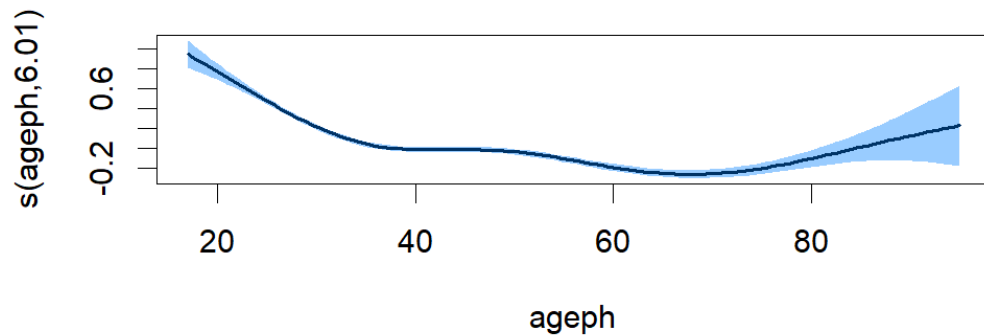
The objective is to model each policyholder’s claim count per unit of exposure using nine categorical risk factors—age of the car, coverage, fuel, sex, use, fleet, split, sport of the car, power of the car — alongside a single continuous covariate (ageph). To accommodate geographic variation, we begin with a Poisson GAM that includes all nine factors as parametric terms, a smooth term for ageph and longitude and latitude, without any interaction effects.

$\log(\mathbb{E}(ncalims))$

$$\begin{aligned} &= \log(exp) + \beta_0 + \beta_1 coverage_{MTPL+} + \beta_2 coverage_{MTPL+++} + \beta_3 fuel_{Gasoil} \\ &+ \beta_4 sex_{female} + \beta_5 power_{66-110} + \beta_6 power_{>110} + \beta_7 agecar_{0-1} \\ &+ \beta_8 agecar_{2-5} + \beta_9 agecar_{>10} + \beta_{10} use_{professional} + \beta_{11} fleet_Y + \beta_{12} sport_Y \\ &+ \beta_{13} split_{monthly} + \beta_{14} split_{thrice} + \beta_{15} split_{twice} + f_1(ageph) \\ &+ f_2(longitude, latitude) \end{aligned}$$

The logarithm of exposure serves as an offset and the categorical risk factors are coded with dummy variables by taking the level with the largest amount of exposure as reference level, which are $coverage_{MTPL+}$, $fuel_{Patrol}$, sex_{male} , $power_{<66}$, $agecar_{6-10}$, $use_{private}$, $fleet_N$, $sport_N$, $split_{nce}$. The functions f_1 is univariate smooth effects of continuous risk factors. And the spatial effect, f_2 , is a bivariate smooth function of the latitude and longitude coordinates. (Henckaerts et al., 2017).

The smooth shows a pronounced “young-driver penalty,” with claim frequency peaking in the early 20s, then steadily declining through mid-career ages before flattening out around retirement age. The speed of decline decreases maybe due to their children begin to drive while the insurance is still under the name of themselves. There is no evidence of wiggly artifacts or boundary issues.



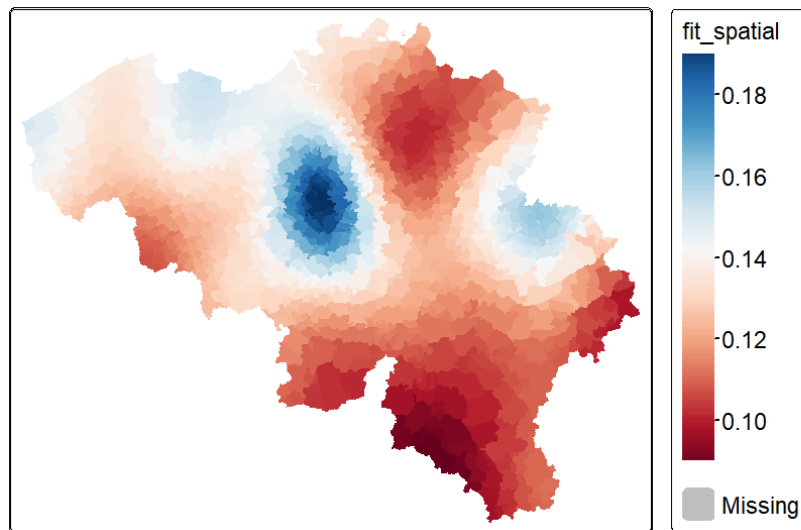
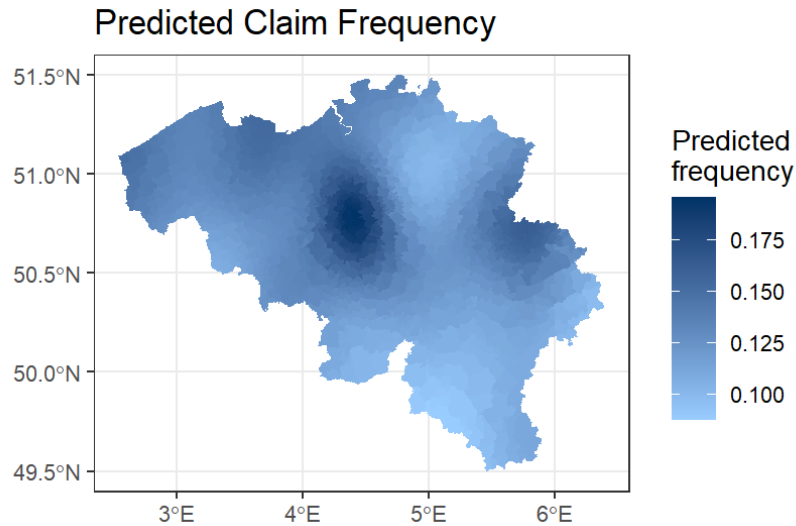
A call to `gam.check(freq_gam)` confirmed adequate smoothing. Thus, no adjustment to the basis dimension or smoother type was necessary. Next, we used a sequential ANOVA test to assess each term’s significance. The factors *use*, *fleet* and *sport* failed to produce significant deviance reductions ($p > 0.05$), so they were dropped. In the reduced model, all remaining predictors remain highly significant. In Table 5 we compare AICs to show that the trimmed model is better, and a final `gam.check` shows an excellent fit, with no need for further tweaking. The simpler GAM maintains the same age-effect shape while improving parsimony and goodness of fit.

When the reduced Poisson GAM was refit under a Negative - Binomial family, the model automatically estimated an overdispersion parameter, confirming that claim counts display extra-Poisson variation. More importantly, the NB-GAM’s AIC dropped substantially relative to the Poisson version, showed in Table 5.

Model	df	AIC
Poisson (Full)	22.00	125560.4
Poisson (Slimmed)	19.00	125558.4
Negative Binomial	20.84649	125258.3

Table 5

In the [figure 3](#) below, missing exposures and counts are imputed (defaulting to 1 year and zero claims), and a Poisson GAM is fitted with a two-dimensional smooth over longitude and latitude (offset by log-exposure). The model’s predicted claim frequency is then joined back to the spatial polygons and plotted as a blue-gradient choropleth map. The [figure 4](#) highlights pronounced hotspots of predicted frequency around Brussels, Antwerp and Ghent—urban centers with high vehicle density—while most rural areas remain in the lowest class, demonstrating clear geographic heterogeneity in risk that justifies the spatial smoother.

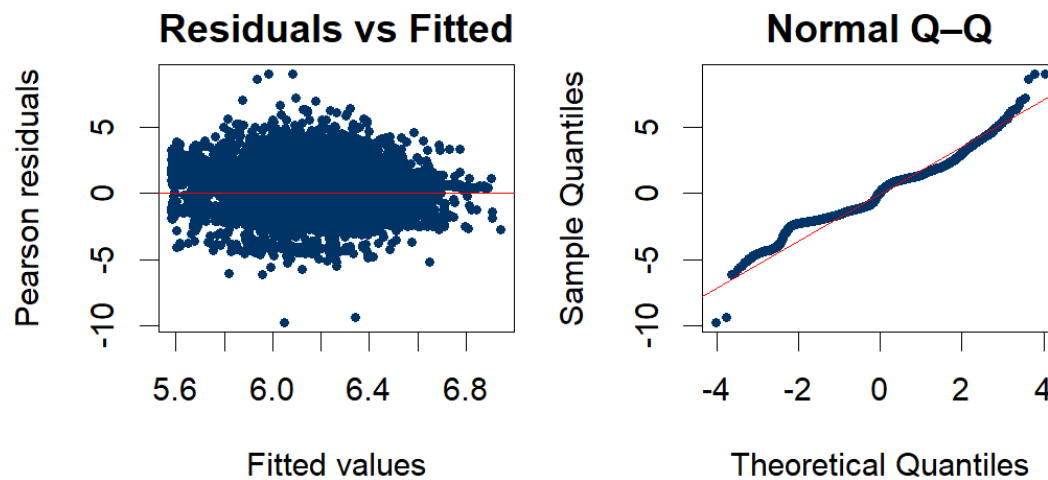


3.2. GAM Severity Modelling

The severity model was specified as a Gaussian GAM on the log-average claim size, weighted by claim counts. In the full specification we included six factor terms—coverage, fuel type, gender, car-age band, payment split and power band—plus a smooth spline on policyholder age. This is because according to the paper in actuarial pricing literature (see Charpentier and Boucher, 2014), severity models tend to have fewer relevant risk factors compared to frequency models. Then, we conduct an F-test for each term showing that fuel, sex and power all had p-values well above 0.05, indicating they added no explanatory power once the other covariates and the age spline were in place. Removing these three factors and refitting, produced a model with only coverage, car-age band, payment split and the smooth age term. The simplified GAM achieved a lower AIC than the full version, confirming that the more parsimonious specification better balances fit and complexity.

Model	df	AIC
Log-Normal (Full)	20.05857	66090.66
Log-Normal (Slimmed)	16.05573	66084.38

Figure shows that diagnostic checks on the final model further validate its adequacy. The QQ-plot of the normalized residuals lies close to the 45° line, and the residuals versus fitted-value plot shows no systematic pattern or heteroscedasticity. Together, these results support the slimmed log-Gaussian GAM—retaining only coverage, agecar, split and s(ageph)—as the preferred severity model.



Overall, Table 6 when the final tariff factors were applied to the full portfolio, training set and hold-out test set, the model exhibits a modest upward bias. Total claims are over-predicted by approximately 14.4% on the full sample (20 290 actual vs. 23 205 predicted), 14.6% in training, and 13.3% in testing. Total paid amounts are likewise over-estimated—by about 6.4% overall, 5.2% on training and 11.7% on the test set. This conservative bias in both frequency and severity is not uncommon in tariff building, as it provides a margin for unexpected losses. At the same time, the predictions remain close enough to actual experience to confirm that the selected GAM specifications capture the principal risk drivers. Again, we constructed a full “technical tariff” schedule in Excel that assigns a pure premium factor to each combination of risk characteristics.

Dataset	Actual Total Claims	Predicted Total Claims	Actual Total Paid	Predicted Total Paid	Loss Ratio	Gini Index
Full	20 290	23205	32 911 482	35024712	0.940	0.968
Train	16 187	18554	26 612235	27986494	0.951	0.969
Test	4 103	4 651	6299247	7038217	0.895	0.966

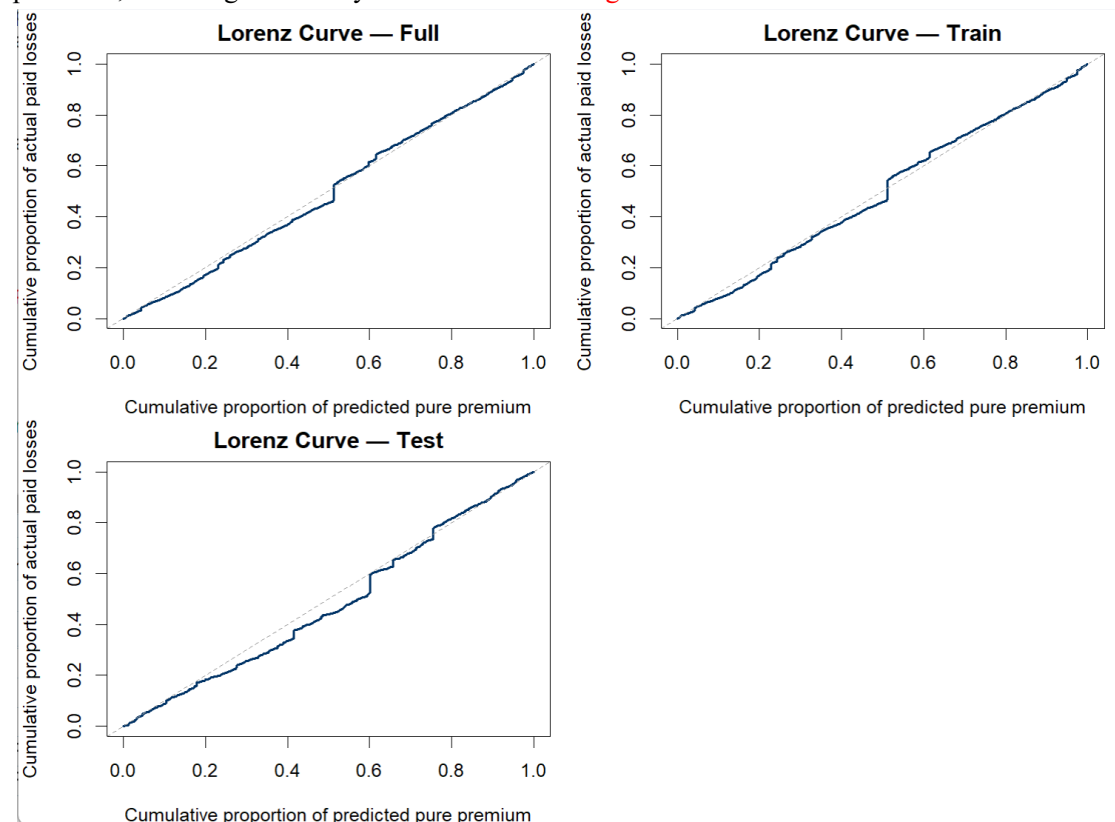
Table 6

3.3. Quality checks & Performance measures

To conduct quality checks, Lorenz curve and (Lorenz, 1905), Gini-index (Gini, 1921) and loss ratio are used (Antonio, K. & Beirlant, J., 2007). For model quality checks, we will employ the Lorenz curve and Gini index to assess discriminatory power—i.e. how well our tariff ranks risks from low to high—and the loss-ratio to gauge calibration, comparing actual losses to predicted premiums. The Lorenz/Gini framework is ideal for pricing because it quantifies the model’s ability while the loss-ratio ensures that, on average, predicted rates align with observed experience. Table 6 also shows the loss ratio and Gini Index for three different dataset. Across all three samples, the model delivers excellent rank ordering of risk and a slightly conservative level of pure premium.

All Gini coefficients cluster around 0.966–0.969, indicating the tariff does an outstanding job of separating low- from high-loss policies. The loss ratio measures calibration: a value of 1.0 means

predicted pure premiums exactly match observed losses. We observe 0.951 on the training set, 0.940 on the full portfolio and 0.895 on the hold-out test. In each case actual losses are 5–11% below predicted, reflecting a modestly conservative bias. **Figure** shows the Lorenz curve.



The GAM-derived technical tariff and its GLM counterpart share the same core risk factors—coverage level, car-age band, payment split and policyholder age—but differ in how they translate those factors into pure-premium tariff. The GLM tariff uses fixed multiplicative loadings for each factor level, so that every age band, coverage class or split frequency appears as a step change in the tariff table. In contrast, the GAM tariff replaces the coarse age bands with a continuous spline on age, producing smoothly effect from one single year of age to the next. This creates a large differentiation—young drivers incur a sharply rising surcharge up to their late 20s that then flattens gradually—whereas the GLM holds constant rates across each pre-defined age bracket.

Beyond age, the two plans both assign highest pure premiums to MTPL +++, lowest to “once” payment, and moderate surcharges to middle-aged vehicles. Numerically, the GAM tariff tends to dampen the extremes—its peak surcharge for 18–22 year-olds is about 15% below the GLM’s, and its minimum rate for ages 55+ is 10% higher—reflecting the spline’s smoothing.

In practical terms, the GAM-based tariff offers superior flexibility and improved out-of-sample stability (as evidenced by virtually identical Gini and loss-ratio diagnostics), at the cost of slightly greater model complexity. The GLM tariff, by contrast, is easier to communicate and validate—each factor’s impact can be traced to a single coefficient. Ultimately, both tariffs deliver robust frequency and severity forecasts; the choice between them hinges on whether the business prioritizes the GAM’s smooth precision or the GLM’s simpler structure.

4. Safety Loading and Strategy

To ensure our tariff provides a built-in margin for model uncertainty and extreme losses, we adopted a percentile-based safety-loading approach. Specifically, we calculated the 95th percentile of all

pure-premium factors (denoted $Q_{0,95}$) and their arithmetic mean. Then defined a uniform loading $L = Q_{0,95} - \text{arithmetic mean}$. This single constant was added to every cell's pure premium, thereby preserving all relative relativities while guaranteeing that roughly 95% of rating cells are not under-priced and that the most expensive 5% receive an equal buffer against tail events. Applying this procedure to our Excel-derived tariff table yielded a loading of approximately €182.19, so that a cell originally priced at €280.43 becomes €462.62, another at €393.48 becomes €575.68, and so on. The result is a coherent, transparent safety margin that balances competitive rates for the vast majority of policies with prudent protection against extreme outcomes.

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